

# Generation Interconnection Combined Feasibility / System Impact Study Report

## For Queue Project Q19-1357

Olivehurst 100 kV  
June 28, 2021

### 1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1357. The Interconnection Customer proposes a 130.2 MW solar photovoltaic facility interconnecting at the Olivehurst 100 kV substation in the Ohio area. The request was received on December 13, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 419 higher-queued active requests and their assigned network upgrades.

### 2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 10 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$15,833,000.

### 3. Network Impacts

#### 3.1 Generator Deliverability and Single Contingency Analysis

| Monitored Facility                     | Contingency                           | Loading (%) | Rating (MVA) | DFAX (%) |
|----------------------------------------|---------------------------------------|-------------|--------------|----------|
| Beechwood-Simons Substation 100 KV     | Lagon Crk-Eighteenth St 100 KV        | 128.7       | 127          | 54.1     |
| Beechwood-Riverplace 100 KV            | Eighteenth St-Lagoon Crk 100 KV       | 124.6       | 151          | 54.1     |
| Carbona-Sierra Linda 1704 100 KV line  | Lagoon Creek-Eighteenth St 100 KV     | 120.1       | 138          | 45.9     |
| Salineville-Lagoon Creek 100 KV        | Eighteenth St-Lagoon Crk 100 KV       | 117.5       | 176          | 54.1     |
| Sierra Linda 1704-Ephrata 100 KV       | Lagoon Creek-Eighteenth Street 100 KV | 111.4       | 165          | 45.9     |
| Ephrata-Olivehurst 100 KV              | Eighteenth Street-Lagoon Crk 100 KV   | 109.1       | 185          | 45.9     |
| Lagoon Creek-Eighteenth St 100 KV Line | Ephrata-Olivehurst 100 KV Line        | 108.2       | 230          | 61.4     |
| Lagoon Creek-Olivehurst 100 KV Line    | Olivehurst-Ephrata 100 KV             | 106.7       | 172          | 100.0    |
| Carbona-MareIsland 100 KV Line         | Lagoon Crk-Eighteenth St 100 KV line  | 105.3       | 142          | 45.9     |
| Salineville-Riverplace 100 KV          | Eighteenth St-Lagoon Crk 100 KV       | 105.1       | 188          | 54.1     |

#### 3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

#### 3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

### 4. Required Network Upgrades and Cost Estimates

| Facility                              | Required Upgrade                            | Cost Estimate | Allocation to Project |
|---------------------------------------|---------------------------------------------|---------------|-----------------------|
| Beechwood-Simmons 100 KV Line         | Rebuild line with higher-capacity conductor | \$3,413,000   | 29% (\$981,000)       |
| Beechwood-Riverplace 100 KV           | Rebuild line with higher-capacity conductor | \$20,343,000  | 29% (\$5,849,000)     |
| Carbona-Sierra Linda 1704 100 KV line | Rebuild line with higher-capacity conductor | \$4,053,000   | 32% (\$1,291,000)     |
| Saineville-Lagoon Crk 100 KV          | Rebuild line with higher-capacity conductor | \$5,307,000   | 29% (\$1,526,000)     |
| Ephrata-Sierra Linda 1704 100 KV Line | Rebuild line with higher-capacity conductor | \$6,351,000   | 32% (\$2,023,000)     |
| Olivehurst-Ehrata 100 KV              | Reconductor line                            | \$2,885,000   | 32% (\$919,000)       |
| Eighteenth St-Lagoon Crk 100 KV line  | Reconductor line                            | \$2,104,000   | 30% (\$633,000)       |
| Lagoon Creek-Olivehurst 100 KV Line   | Reconductor line                            | \$1,594,000   | 67% (\$1,064,000)     |
| Carbona-MareIsland 100 KV Line        | Reconductor line                            | \$1,909,000   | 32% (\$608,000)       |
| Salineville-Riverplace 100 KV         | Reconductor line                            | \$3,267,000   | 29% (\$939,000)       |
|                                       | Total allocated to project                  |               | \$15,833,000          |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.